

Partial Result: A Uniform Tail Property-(H) Obstruction

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Source Problem

The source is Valentin Ferenczi, Jordi Lopez-Abad, Behrang Mbombo, and Sebastien Todorcevic, *Local Banach-space dichotomies and ergodic spaces*, arXiv:2005.06458. Question 6.17 appears on PDF page 60 and asks:

1. Does every non-Hilbertian space have a non-Hilbertian subspace with a Schauder basis?
2. Does every non-Hilbertian space with unconditional FDD have a non-Hilbertian subspace with an unconditional basis?

This packet addresses the second question.

case (see [49] for these notions). For non-Hilbertian examples of HAPpy spaces with a symmetric basis (and therefore also non asymptotically Hilbertian), see [30].

Theorem 6.15 naturally opens the following two questions (the first one had already been asked by Pełczyński [43]):

Question 6.17.

1. Does every non-Hilbertian space have a non-Hilbertian subspace with a Schauder basis?
2. Does every non-Hilbertian space with unconditional FDD have a non-Hilbertian subspace with an unconditional basis?

Proof of Theorem 6.15.

1. Let X be a d_2 -minimal space, and fix λ_0 as given by Lemma 6.14 for X . Build an FDD $(F_n)_{n \in \mathbb{N}}$ of a subspace of X , along with a decreasing sequence $(Y_n)_{n \in \mathbb{N}}$ of finite codimensional subspaces of X , by induction as follows. Let $Y_0 = X$. The

Definitions

Let X be a Banach space with an unconditional finite-dimensional decomposition (FDD) $(E_n)_{n=1}^\infty$. A normalized block sequence of (E_n) means a sequence of unit vectors with successive finite supports with respect to the FDD.

Definition 1 (uniform tail property-(H) failure along an FDD). *We say that (E_n) has uniform tail property-(H) failure if there is $\lambda \geq 1$ such that for every $N \in \mathbb{N}$ and every $M \geq 1$, there are integers $N \leq r \leq s$ and a finite-dimensional subspace*

$$G \subseteq E_r \oplus E_{r+1} \oplus \cdots \oplus E_s$$

which has a normalized λ -unconditional basis and satisfies

$$d_{\text{BM}}(G, \ell_2^{\dim G}) > M.$$

This condition is a tail version of failing Pisier's property (H) with a fixed unconditionality constant. It is exactly the finite-dimensional supply used in the source paper's proof of Theorem 6.15 for d_2 -minimal spaces, but here it is stated as an explicit hypothesis rather than derived from minimality among non-Hilbertian spaces.

Idea of the Proof

If some normalized block sequence of the FDD already spans a non-Hilbertian space, the unconditionality of the FDD gives the desired unconditional basis. Otherwise every normalized block sequence is Hilbertian. A standard gliding hump argument then upgrades this to a uniform statement: all normalized block sequences are C -equivalent to the unit vector basis of ℓ_2 . The tail witnesses supply successive finite-dimensional blocks whose internal bases are uniformly unconditional but whose Hilbertian distortion tends to infinity. The uniform ℓ_2 behavior of normalized block vectors controls sign changes between the blocks, while the internal unconditionality controls sign changes inside each block. Concatenating the block bases therefore gives an unconditional basis. Since the blocks have unbounded Banach-Mazur distance from Euclidean spaces, their closed span cannot be Hilbertian.

Theorem 1. *Let X be a Banach space with an unconditional FDD (E_n) . If (E_n) has uniform tail property-(H) failure, then X contains a non-Hilbertian subspace with an unconditional basis.*

Proof. Let K be an unconditionality constant for the FDD. If there is a normalized block sequence (u_j) of (E_n) whose closed span is non-Hilbertian, then (u_j) is an unconditional basic sequence, with constant bounded in terms of K , and we are done.

Assume therefore that every normalized block sequence spans a Hilbertian subspace. We first record the usual uniformity consequence. There is a constant C such that every normalized block sequence of (E_n) is C -equivalent to the unit vector basis of ℓ_2 . Indeed, if no such constant existed, then for every tail and every integer m one could choose a finite normalized block sequence in that tail which is not m -equivalent to the Euclidean basis of the same length. Choosing these finite sequences successively and concatenating them gives a normalized block sequence. By the current assumption its span is Hilbertian; since the sequence is unconditional, the standard Hilbert-space fact for normalized unconditional basic sequences implies that it is equivalent to the ℓ_2 basis. This contradicts the successively chosen finite blocks with constants tending to infinity.

Let λ witness uniform tail property-(H) failure. Inductively choose successive finite-dimensional block subspaces G_k , each supported after the previous one, such that G_k has a normalized λ -unconditional basis

$$(x_i)_{i_k \leq i < i_{k+1}}$$

and $d_{\text{BM}}(G_k, \ell_2^{\dim G_k}) \geq k$. We claim that the concatenated sequence $(x_i)_{i=0}^\infty$ is unconditional.

Fix finitely supported scalars (a_i) and signs (ε_i) . For each k , put

$$u_k = \sum_{i_k \leq i < i_{k+1}} a_i x_i, \quad v_k = \sum_{i_k \leq i < i_{k+1}} \varepsilon_i a_i x_i.$$

Let $b_k = \|u_k\|$ and $c_k = \|v_k\|$. By the λ -unconditionality of the basis inside G_k ,

$$c_k \leq \lambda b_k.$$

When $b_k \neq 0$, write $y_k = u_k/b_k$, and when $c_k \neq 0$, write $z_k = v_k/c_k$. Omitting zero terms, (y_k) and (z_k) are normalized block sequences of the original FDD. Hence they are C -equivalent to the ℓ_2 basis. Therefore

$$\left\| \sum_i \varepsilon_i a_i x_i \right\| = \left\| \sum_k c_k z_k \right\| \leq C \left(\sum_k c_k^2 \right)^{1/2} \leq \lambda C \left(\sum_k b_k^2 \right)^{1/2} \leq \lambda C^2 \left\| \sum_k b_k y_k \right\| = \lambda C^2 \left\| \sum_i a_i x_i \right\|.$$

Thus (x_i) is unconditional.

Let $Y = [x_i : i \in \mathbb{N}]$. Since each G_k is a finite-dimensional subspace of Y and $d_{\text{BM}}(G_k, \ell_2^{\dim G_k}) \rightarrow \infty$, the space Y cannot be isomorphic to a Hilbert space: a Hilbertian isomorphism with constant D would make every finite-dimensional subspace of Y D -isomorphic to a Euclidean space. Hence Y is a non-Hilbertian subspace of X with an unconditional basis. $\square$

Consequence for Question 6.17(2)

This is a partial positive result for Question 6.17(2). It shows that any counterexample with an unconditional FDD must be much closer to Hilbert space than a naive non-Hilbertian UFDD space: either it has no non-Hilbertian block subspace, or every fixed unconditionality constant becomes Hilbertian on far enough tails. In particular, the obstruction to the source question is not merely the lack of bases inside the finite-dimensional FDD blocks; it must also prevent uniformly unconditional, increasingly non-Hilbertian finite blocks from appearing arbitrarily far out.

Novelty and Search Notes

The cheap run indexes were searched for arXiv:2005.06458, the title phrase, “unconditional FDD”, “non-Hilbertian”, “property (H)”, “MNH”, and related ergodicity keywords. The only relevant existing row was a separate literature-implied answer about Schlumprecht space ergodicity; no packet for Question 6.17 or this tail-property route was found. Exact web searches on June 14, 2026 for the Question 6.17 wording and for property-(H)/UFDD variants did not surface a separate later answer.

This theorem is not a full solution of Question 6.17(2). It abstracts and extends the block concatenation mechanism used in the source paper’s proof of Theorem 6.15 from the special d_2 -minimal setting to spaces where the needed finite-dimensional tail witnesses are assumed directly.

Human Review Recommendation

Review the uniformity claim for normalized block sequences and the final concatenation estimate. The proof is short and should be checked mainly for whether the stated tail witness hypothesis is the right level of generality for a useful partial result.

References

- [1] V. Ferenczi, J. Lopez-Abad, B. Mbombo, and S. Todorcevic, *Local Banach-space dichotomies and ergodic spaces*, arXiv:2005.06458.